

AssistedDS: Benchmarking How External Domain Knowledge Assists LLMs in Automated Data Science

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Abstract

Large language models (LLMs) have advanced the automation of data science workflows. Yet it remains unclear whether they can critically leverage external domain knowledge as human data scientists do in practice. To answer this question, we introduce AssistedDS (Assisted Data Science), a benchmark designed to systematically evaluate how LLMs handle domain knowledge in tabular prediction tasks. AssistedDS features both synthetic datasets with explicitly known generative mechanisms and real-world Kaggle competitions, each accompanied by curated bundles of helpful and adversarial documents. These documents provide domain-specific insights into data cleaning, feature engineering, and model selection. We assess state-of-the-art LLMs on their ability to discern and apply beneficial versus harmful domain knowledge, evaluating submission validity, information recall, and predictive performance. Our results demonstrate three key findings: (1) LLMs frequently exhibit an uncritical adoption of provided information, significantly impairing their predictive performance when adversarial content is introduced, (2) helpful guidance is often insufficient to counteract the negative influence of adversarial information, and (3) in Kaggle datasets, LLMs often make errors in handling time-series data, applying consistent feature engineering across different folds, and interpreting categorical variables correctly. These findings highlight a substantial gap in current models’ ability to critically evaluate and leverage expert knowledge, underscoring an essential research direction for developing more robust, knowledge-aware automated data science systems.

1 Introduction

Recent advancements in large language models (LLMs) have led to significant progress in automating data science workflows. LLMs such as GPT-

4 (Achiam et al., 2023) and Claude (Anthropic, 2025) have shown remarkable capabilities in generating code and performing machine learning tasks, enabling end-to-end automation of many routine data analysis procedures (Grosnit et al., 2024; Hong et al., 2025; Jiang et al., 2025; Liang et al., 2025).

Despite these capabilities, a critical aspect of LLMs for real-world data science remains underexplored: the effective integration and critical evaluation of external domain knowledge. In practice, data scientists do not simply run standard algorithms—they routinely incorporate domain-specific knowledge, solicit feedback from colleagues, and weigh multiple sources of information before finalizing decisions (Mao et al., 2019; Zhang et al., 2020). Such processes are essential for handling the complexities and ambiguities inherent in real-world data tasks.

However, current research on LLM-driven data science has largely focused on code generation and pipeline execution (Li et al., 2024b; Jiang et al., 2025), neglecting the crucial role of external expert knowledge in practical applications. Existing evaluation benchmarks (Chan et al., 2025; Jing et al., 2025; Zhang et al., 2025) rarely measure how LLMs process, filter, or critically adopt domain knowledge, especially when such information may be misleading or adversarial. This raises fundamental questions: Can LLMs distinguish between helpful and harmful external domain knowledge in data science workflows? Or do they simply adopt all information, risking degraded performance when exposed to adversarial guidance?

In this paper, we introduce **AssistedDS** (Assisted Data Science), a benchmark designed specifically to measure the capability of LLMs to leverage domain knowledge effectively and critically for data science. AssistedDS evaluates LLM performance on tabular prediction tasks augmented with diverse domain knowledge bundles, which include both helpful domain insights and deliberately ad-

*Equal distribution.

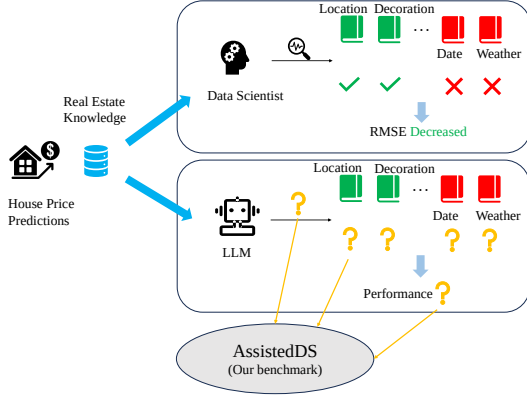


Figure 1: How AssistedDS establishes its unique position. **Top:** A human data scientist analyses data, and often critically filters external domain knowledge. This means accepting helpful hints (green) and rejecting adversarial ones (red), which leads to demonstrably better results. **Bottom:** An LLM can automate the same analytical steps, but its ability to judge the quality of domain knowledge is unknown. AssistedDS evaluates whether an LLM, like its human counterpart, can identify and leverage helpful information while resisting harmful advice in automated data science workflow

versarial information. Each task within AssistedDS comes with synthetic datasets that have ground-truth generative mechanisms, enabling precise characterization of helpful versus adversarial domain knowledge. We complement these synthetic experiments with real-world Kaggle datasets (Ferguson et al., 2016; Cukierski, 2014; Montoya et al., 2016; Bossan et al., 2015; FlorianKnauer and Cukierski, 2015), where helpfulness is defined by Kaggle-hosted highly-rated expert notebooks and adversarial examples are constructed from low-quality notebooks.

Our experiments reveal three critical findings. First, LLMs consistently demonstrate an uncritical tendency to follow provided information, even in the presence of adversarial cues. Our experiments on synthetic datasets show that even GPT-o4-mini fails to filter out 60% of adversarial hints, and, in the worst case, GPT-4o-mini blindly follows every adversarial hints. All these blind following of adversarial hints leads to substantial performance declines, as much as -30% . Second, prompts indicating helpful versus adversarial content provides only limited improvement, highlighting a fundamental limitation in current models’ ability to critically evaluate external sources. Third, in addition to findings in synthetic datasets, we find that LLM’s use of domain knowledge for real-world data such

as Kaggle has new issues like deficiencies in handling time-series data, leading to enormous failures code execution. For instance, in the Bike Sharing competition, LLMs generated 58 “unknown category” failures (25.78% of all errors) when asked to predict 2012 outcomes from 2011 data, with 38 of those errors (70.7%) explicitly mentioning “2012.” This clearly shows that the models treat years as isolated categories instead of recognizing them as points on a continuous timeline.

These results underscore a crucial limitation in current LLM-driven automation: the lack of critical reasoning about external domain knowledge. Recognizing this shortcoming, we argue that future research must focus on developing LLM capabilities for critical assessment and selective integration of external domain knowledge. By highlighting these persistent weaknesses, AssistedDS lays foundational groundwork for robust, knowledge-aware automated data science.

Our main contributions are summarized below.

- We propose **AssistedDS**, the first benchmark to evaluate LLMs’ ability to critically integrate and filter external domain knowledge in automated data science workflows, including both synthetic and real-world datasets.
- We curate controlled bundles of helpful and adversarial domain knowledge, enabling assessment of LLM’s robustness under mixed and ambiguous informational environments.
- We present empirical evidence in synthetic datasets that state-of-the-art LLMs frequently adopt external knowledge uncritically, suffering substantial performance degradation in the presence of adversarial information.
- We demonstrate that these vulnerabilities persist in real-world data science tasks and found that LLMs often fail in time-series handling, feature engineering consistency, and categorical data interpretation, underscoring the need for knowledge-aware automation.

1.1 Related Work

Automated Data Science with LLMs. Automating data science workflows using LLMs has seen considerable interest in recent research. AutoKaggle (Li et al., 2024b) structures the data science process into clearly defined stages, employing specialized agents for iterative coding and debugging. Agent K (Grosnit et al., 2024) frames data

science tasks as a Markov Decision Process, enabling agent-based interactions that include both internal reasoning and external actions. Similarly, Data Interpreter (Hong et al., 2025) models the data science workflow dynamically as a hierarchical graph, optimizing performance by iteratively refining task dependencies. SPIO (Seo et al., 2025) uses a central planner to explore multiple predictive strategies, either selecting a single best strategy or forming ensembles to achieve robust performance. Despite these advances, current automated workflows rarely evaluate the integration and impact of external domain knowledge.

Evaluation Benchmarks. Benchmarks like MLE-bench (Chan et al., 2025) focus on evaluating end-to-end predictive performance of machine learning pipelines. DA-Bench (Hu et al., 2024) and DA-code (Huang et al., 2024) benchmark code generation in data science. DS Bench (Jing et al., 2025) and DataSciBench (Zhang et al., 2025) expand these evaluations beyond mere predictive accuracy, probing reasoning capabilities through curated tasks that involve interpretation and logical assessment of data. However, these benchmarks do not explicitly examine how well LLMs incorporate external guidance, particularly when such information varies in quality or reliability.

Reasoning and Critical Evaluation in LLMs. Recent studies addressing reasoning capabilities in LLM-driven workflows emphasize iterative tree-based search methods. AIDE (Jiang et al., 2025), SELA (Chi et al., 2024), and I-MCTS (Liang et al., 2025) leverage structured search strategies to iteratively refine solutions, suggesting improved reasoning about complex tasks. DS-Agent (Guo et al., 2024) retrieves high-quality external knowledge sources, such as Kaggle notebooks from experts, to enable case-based reasoning. Yet, these approaches primarily demonstrate positive use cases and lack systematic analysis of LLMs’ vulnerability to adversarial or misleading external information.

Retrieval-Augmented Generation and Long Context Models. To reduce hallucination and improve factual accuracy, recent research has employed retrieval-augmented generation (RAG) (Lewis et al., 2020; Khandelwal et al., 2019) and long context (LC) modeling (Chen et al., 2023; Wang et al., 2024). LC methods generally outperform RAG for structured, dense contexts, effectively handling well-defined information such as notebook tutorials (Li et al., 2024a). Given the structured nature of external data science knowl-

edge, our study adopts the LC approach by directly incorporating comprehensive external context into LLM prompts. Our work addresses crucial gaps in the existing literature by benchmarking the capability of LLMs to integrate external domain expertise critically.

Our work contributes by examining the critical capacity of LLMs to leverage external domain knowledge for automated data science, evaluating their performance against both beneficial and adversarial information. This focused evaluation sets our work apart from existing literature, highlighting both the strengths and fundamental limitations of current automated data science approaches.

2 Benchmark Curation

Our benchmark, **AssistedDS**, evaluates how effectively and critically LLMs leverage domain knowledge in automated data science workflows. An overview of our benchmark framework is illustrated in Figure 2. We consider synthetic datasets to have control on domain knowledge, and Kaggle datasets to reflect real-world complexities on domain knowledge. For synthetic datasets we generate dataset with transparent generating mechanism, and then where we can define and inject “helpful” hints that boost performance and “adversarial” hints that can cause performance decline. For Kaggle datasets we draw on real-world problems from Kaggle competitions, pairing each dataset with high-quality community notebooks that embody genuine domain knowledge through feature engineering, data cleaning, and analysis. To mirror realistic scenario for LLM’s access to domain knowledge, we then designed tasks, which are different bundles of helpful and adversarial domain knowledge, paired with certain prompt template to let LLM generate an end-to-end code for submission.csv. Then we evaluate based on the code script generated by the LLM. Below, we elaborate more on why and how we prepare the datasets, domain knowledge, and evaluation metrics, and in Section 3.1 we describe the tasks as experimental settings.

2.1 Datasets

Our benchmark encompasses both synthetic and real-world datasets. The synthetic datasets give us full control over the domain knowledge: we construct feature-label relationships to produce helpful hints (which should improve predictive performance) and adversarial hints (which should de-

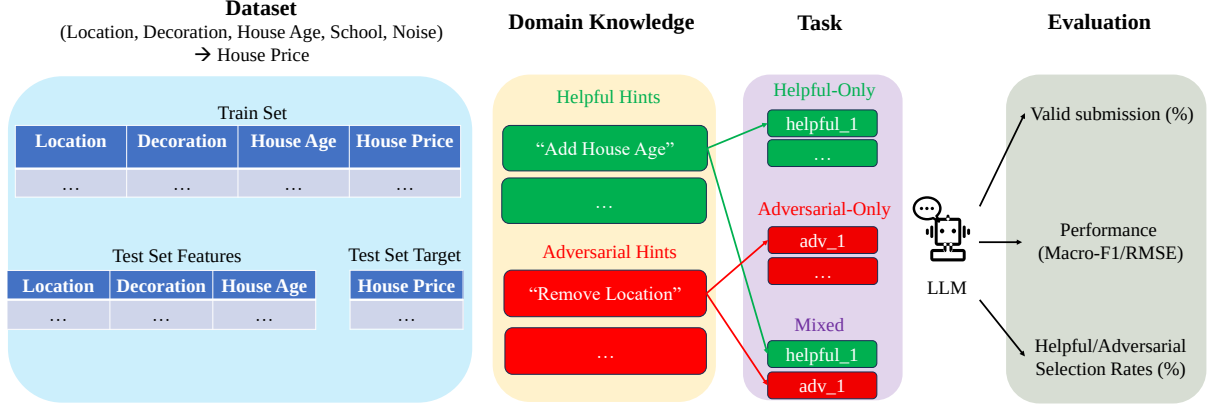


Figure 2: Overview of the AssistedDS Benchmark. **Left: Dataset.** We use datasets (synthetic or real-world) with defined features (e.g., Location, Decoration, House Age, School Quality) and clear train/test splits for tabular prediction tasks. **Center-left: Domain Knowledge.** For each dataset, we curate bundles of domain knowledge: helpful hints (e.g., “Add School Quality”) and adversarial hints (e.g., “Remove Location”), reflecting realistic expert guidance and potential pitfalls. **Center: Task Bundles.** Hints are grouped into tasks: Helpful-Only, Adversarial-Only, Mixed, and so on. Each represents different real-world information environments. **Center-right: LLM Execution.** An LLM receives both the data and bundled hints, together with the prompt, and generates a end-to-end code script. **Right: Evaluation.** We assess each model’s generated code by (i) valid submission rate, (ii) predictive performance (Macro-F1 or RMSE), and (iii) the rates at which helpful or adversarial hints are actually adopted in the generated code.

grade them). The real-world collection consists of competitions drawn from Kaggle, each accompanied by community notebooks rated by us that capture authentic domain expertise—ranging from data preprocessing to feature manipulation. In the sections that follow, we describe how synthetic datasets and Kaggle datasets are curated.

Synthetic Datasets. We curate 10 synthetic tabular datasets, encompassing 6 classification and 4 regression tasks. Each dataset is generated with a clearly defined data-generating process involving realistic domain-specific features, nonlinear relationships, interactions, noise, and intentional data imperfections such as missing values and outliers. The datasets cover diverse application domains including Game Revenue, Real Estate, Second-hand Goods, Power Generation, Diabetes, Machine Failure, Haircut Rate, Wine Quality, Song Popularity, and Housekeeping. For each dataset, we provide standard training/test splits, along with additional files for new or manipulated features for domain knowledge preparation. A summary of synthetic datasets we curated is in Appendix A.

Illustrative Example: Song Popularity We illustrate our synthetic dataset generation using the “Song Popularity” domain. Domain-relevant features are first specified, followed by the response variable (*Popularity Class*) capturing realistic rela-

tionships and stochasticity. Specifically, the *Popularity Score* is generated from a linear combination of domain-specific features such as Artist Fame (positive effect), Genre (pop and hip-hop positively affect popularity, other genres negatively), Danceability (strong quadratic positive effect), Tempo in BPM (positive), high Social Media Hype (positive binary effect), presence of Featuring Artist interacting positively with Artist Fame, squared Lyric Sentiment (negative quadratic effect), Music Video Budget (positive), and Album Position (negative). A logistic noise term is added to mimic real-world stochasticity. This continuous score is then discretized into three classes: *Unpopular*, *Ordinary*, and *Hot*, based on predetermined thresholds. The details of how we generated this dataset is in Appendix B. We then hold out two features, Music Video Budget and Tempo BPM, for domain knowledge creation, as explained in Section 2.2.1.

Kaggle Datasets. To capture real-world challenges in both datasets and domain knowledge that synthetic datasets could not exhibit, we include Kaggle datasets as a complement to synthetic datasets in our benchmark. Importantly, we do not simply replicate Kaggle competitions. Instead, we carefully subsample the released datasets to reduce overlap with tasks that may have appeared in LLMs’ pretraining, thereby mitigating

the risk of overfitting due to memorization. Beyond data, we also extract and curate community-authored Kaggle notebooks as a source of domain knowledge, which we systematically rate and rank (see Section 2.2.2) to support downstream evaluation. We select five widely participated competitions: Allstate Claims Severity (Ferguson et al., 2016), Bike Sharing Demand (Cukierski, 2014), BNP Paribas Cardif Claims Management (Montoya et al., 2016), Otto Group Product Classification Challenge (Bossan et al., 2015), and Rossmann Store Sales (FlorianKnauer and Cukierski, 2015). For each dataset, we construct a fixed train/test split and release three files: `train.csv`, `test.csv`, and a `test-target.csv` containing identifiers and ground truth labels used exclusively for evaluation.

Detailed dataset generation procedures and examples are provided in Appendix B.

2.2 Domain Knowledge (Hints)

Domain knowledge plays a central role in our benchmark, as it enables us to evaluate how LLMs integrate external information in automated data science workflows. While we introduced the use of domain knowledge in Section 2.1, we now provide further details on its construction and use.

To assess whether LLMs can critically utilize domain knowledge when generating end-to-end code for tabular prediction tasks, we curate two distinct types of domain knowledge: helpful and adversarial. An LLM that critically engages with domain knowledge should selectively incorporate helpful hints while rejecting misleading ones, leading to improved prediction performance. In contrast, an LLM that uses domain knowledge uncritically—absorbing any hint indiscriminately—is likely to suffer degraded performance.

Below, we detail the procedures for preparing domain knowledge in synthetic and Kaggle datasets.

2.2.1 Domain knowledge: Synthetic Datasets

For synthetic datasets, helpful hints truthfully reflect data-generating processes or proven domain knowledge, while adversarial hints intentionally mislead by advocating inappropriate modeling choices. Hints are combined into bundles with controlled compositions to simulate various realistic informational environments.

Take Song Popularity dataset as an example again. Based the generating mechanism, we provide specific hints aimed at helpful or adversarial modeling choices:

Helpful Hints

Helpful Hints encourage the addition of predictive features previously withheld:

- Add feature Music-Video-Budget: Estimated music video production budget (units of 10,000 dollars, continuous, [0, 500]).
- Add feature Tempo-BPM: Tempo of the song (continuous, [60, 200] BPM).

Adversarial Hints

Adversarial Hints mislead by suggesting removal of important predictive features:

- Remove (crucial predictive) feature `in-game-purchases`.
- Remove (crucial predictive) feature `monetization-model-Premium`.

2.2.2 Domain Knowledge: Kaggle Datasets

To reflect real-world domain knowledge, we use the community notebooks (Kaggle, 2025) from Kaggle platform as domain knowledge. These notebooks often give insights like how features should be created to boost performance. To collect high-quality notebooks, we collected the *Code* section of each competition and ranked available notebooks by vote count. We then retrieved the top 50 notebooks for each competition. For each notebook, we extracted source code, number of votes, number of comments and model performance (e.g., MAE, accuracy), and then designed a composite quality score to evaluate the informativeness and structure of each notebook. The score for each notebook is calculated as

$$0.3 \times \mathcal{S} + 0.5 \times \mathcal{V} + 0.2 \times \mathcal{C}, \quad (1)$$

where $\mathcal{S}$ is the normalized log-score, $\mathcal{V}$ is the normalized log-vote count, and $\mathcal{C}$ is the normalized log-comment count for each notebook. Normalization was performed using min-max scaling across all collected notebooks. Based on these scores, we selected the top 4 notebooks as high-quality exemplars and set them as helpful domain knowledge. For adversarial domain knowledge, we take the bottom 4 as low-quality notebooks for each competition, and remove crucial feature in the code to make them adversarial. See details in Appendix C.

2.3 Protocols and Models Used for Evaluation

With the datasets and corresponding domain knowledge in place, we next assess the LLMs’ end-to-end automated data science capabilities. Specifically, we evaluate the Python code generated from a single prompt. Each prompt includes a dataset description, a dataset preview, and a bundle of domain knowledge, as specified in Section 3. In this paper, we evaluate three LLMs from OpenAI: GPT-o4-mini, a cost-effective model that delivers reasoning capabilities, GPT-4o, the flagship model, and GPT-4o-mini, a more compact variant of GPT-4o.

2.4 Evaluation Metrics

- **Valid Submission (%)**: Proportion of runs in which the `submission.csv` is produced without error, and is formatted correctly for evaluation with `test-target.csv`. This is to check code-execution
- **Helpful Selection (%)**: Number of helpful hints used in the generated code / number of helpful hints provided. This is to measure how helpful domain knowledge is used by LLMs under different settings, and the higher the better.
- **Adversarial Selection (%)**: Number of adversarial hints used in the generated code / number of adversarial hint provided. This is to measure how adversarial domain knowledge is used by LLMs under different settings, and the lower the better.
- **Performance**: RMSE for regression and macro-F1 for classification. This is to see how predictive performance would change when different bundles of domain knowledge are provided. We only report this for synthetic datasets. To indicate the change, we also report **Performance Change**, which is the percentile change from the case where no domain knowledge is provided. We adjust it for RMSE so that for both regression and classification positive values indicate improvement and negative values indicate degradation.

For both helpful and adversarial selections, we employ GPT-4.1 model as an evaluator to assess which hints are actually understood and incorporated in the code generated by the LLM.

3 Experimental Studies

To test an LLM’s ability to critically leverage both helpful and adversarial domain knowledge curated

as described in Section 2.2, we design experiments (tasks) using various combinations of helpful and adversarial hints together with different prompt configurations. The overall design principle is to vary both the proportion of helpful versus adversarial hints and the degree to which prompts inform the LLM about the quality of these hints, allowing us to evaluate how LLMs distinguish and utilize domain knowledge under realistic conditions.

3.1 Tasks

For synthetic datasets, we design five tasks:

- **None** This task feeds no domain knowledge to LLMs. Such case would indicate how LLMs perform when no domain knowledge is present for decision-making, which would serve as a baseline performance.
- **Helpful-Only** consists of $n \in \{1, 2\}$ helpful hints only. We use **neutral prompt** here, i.e., we do not provide quality information regarding domain knowledge. This task is to investigate how well LLMs can follow helpful domain knowledge when this is the only source. Ideally this would ensure the best performance if LLMs takes all helpful advice.
- **Adversarial-Only** consists of $n \in \{1, 2\}$ adversarial hints only. We use **neutral prompt** here, i.e., we do not provide quality information regarding domain knowledge. If LLMs just blindly follow adversarial hints in their code, their predictive performance would drop by our design of “adversarial” hint.
- **Mixed** consists of mixed helpful and adversarial hints, with count being Helpful: Adversarial $\in \{1:1, 2:1, 1:2, 2:2\}$. We use **neutral prompt** here, i.e., we do not provide quality information regarding domain knowledge. This task would test LLMs ability to choose between helpful and adversarial hints when they are mixed. Taking only helpful hints in the code would be ideal.
- **Mixed (Misleading)** consists of mixed helpful and adversarial hints, with count being Helpful: Adversarial $\in \{1:1, 2:1, 1:2, 2:2\}$. We use **adversarial prompt** here, i.e., a deliberately misleading prompt telling LLMs that the helpful hint is adversarial and the adversarial hint is helpful. This task would test if

LLMs can actually discern helpful and adversarial hints, or they are just following what the user prompt tells them.

Additional tasks for the Kaggle datasets are provided in Appendix D.

4 Results and Analysis

In this section, we present our findings and analysis on the aforementioned tasks. All experiments are conducted with 5 independent LLM queries per setting. Detailed table for these results are included in Appendix J.

LLMs tend to blindly follow both helpful and adversarial hints Table 1 summarizes the rates at which each model adopts helpful or adversarial hints under different task conditions. For all models, when only helpful hints are given, they almost always incorporate them into their generated code, resulting in consistent adoption rates of 100%. However, when only adversarial hints are provided, models like GPT-o4-mini follow 60% of such hints, while GPT-4o and GPT-4o-mini follow nearly all (90% and 100%, respectively), indicating an inability to effectively filter adversarial information. In the misleading setting, both GPT-4o and GPT-o4-mini are frequently fooled by misleading prompts, leading to high selection rates of adversarial hints and neglect of helpful ones. GPT-4o-mini likely does not understand the user prompt on quality of domain knowledge, consistently taking all hints even with misleading prompts. Overall, this shows that current LLMs tend to follow provided hints uncritically, regardless of whether the information is beneficial or misleading.

The blind adoption of adversarial hints and misleading prompt leads to substantial declines in performance Figure 3 displays the impact of hint adoption on predictive performance across different models and task settings. As expected, all models exhibit performance gains when following helpful hints. Conversely, the uncritical following of adversarial hints leads to marked performance declines—up to -32% for GPT-4o-mini. Notably, GPT-o4-mini demonstrates some resilience in mixed settings, maintaining positive performance where other models decline. These results underscore a fundamental vulnerability: current LLMs often lack the critical reasoning needed to distinguish beneficial from harmful external knowledge. This limitation poses significant risks

in real-world automated data science workflows, where adversarial or low-quality information may be present. Our findings highlight the urgent need for future research on LLMs that can more robustly evaluate, filter, and reason about domain knowledge, rather than naively accepting all available guidance. Developing such critical assessment capabilities will be crucial for building trustworthy, knowledge-aware automated data science systems.

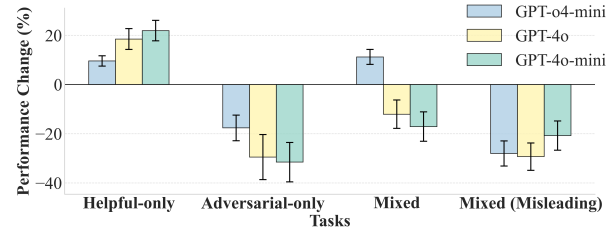


Figure 3: **Performance Change** of LLMs with Different Tasks in Synthetic Datasets. This chart displays the percentage change in model performance relative to the baseline condition (no hint provided). Such change is calculated differently for regression task so that positive value indicates gain in performance. The four tasks represent different bundles of helpful and adversarial hints: "Helpful-only" (only helpful hints provided), "Adversarial-only" (only adversarial hints provided), "Mixed" (both helpful and adversarial hints are provided), and "Mixed (Misleading)" (both helpful and adversarial hints are provided, with intentionally misleading prompts). Each bar is the performance change averaged over settings per task and model. Error bars show standard errors.

Valid submission rates varies by tasks and models Table 2 presents the valid submission rates of models under different task conditions. GPT-o4-mini achieved the highest consistency across tasks, notably excelling in adversarial-only and misleading mixed conditions with rates of 95% and 95.5%, respectively. This is likely because under these two cases GPT-o4-mini only follows adversarial hints which merely involve removing a feature and less likely cause errors in code execution. GPT-4o maintained relatively stable submission rates around 75% to 82%, showing moderate resilience of error. GPT-4o-mini only made submissions below 70% when provided with domain knowledge, indicating that GPT-4o-mini is not able to deal with domain knowledge on dataset manipulation well.

LLMs still tend to blindly follow adversarial domain knowledge under misleading prompts in Kaggle datasets For Kaggle datasets we report

Table 1: Selection rates (%) computed with the total number of selected documents divided by the total number of provided documents across models and tasks for synthetic data

Model	Helpful-only	Adversarial-only	Mixed		Mixed (Misleading)	
	Help	Adv	Help	Adv	Help	Adv
GPT-o4-mini	100.00	60.00	100.00	60.67	2.67	100.00
GPT-4o	100.00	90.67	100.00	97.00	8.00	100.00
GPT-4o-mini	100.00	100.00	100.00	99.33	96.33	100.00

Table 2: Valid submission rates (%) across models and tasks for synthetic data

Model	None	Helpful-only	Adversarial-only	Mixed	Mixed (Misleading)
GPT-o4-mini	84	77	95	83	95.5
GPT-4o	82	75	79	79.5	76.5
GPT-4o-mini	86	60	66	63	60.5

the experimental results in which the LLMs are fed a mixed set of helpful and adversarial notebooks, together with misleading prompt. See Table 17, Table 18 and Table 19 in Appendix J.2 for GPT-4o-mini, GPT-4o and GPT-o4-mini respectively. We observed that the LLMs tend to blindly follow users’ prompts or instructions when selecting adversarial notebooks. This coincides our observation in synthetic datasets.

Failures in temporal, feature, and non-numerical data handling in Kaggle datasets

Our Kaggle experiments reveal that current LLMs exhibit systematic weaknesses across three core aspects of practical data science workflows. First, LLMs struggle with time-series data, frequently treating temporal variables as ordinary categories rather than as points on a timeline. For example, in the Bike Sharing competition, 58 failures (25.78%) were due to “unknown category” errors, with 38 cases (70.7%) explicitly involving the year 2012 when models failed to generalize from 2011 to 2012 data. Date parsing errors were also common, as seen in 10 Rossmann Store failures. Second, LLMs often mishandle feature engineering, failing to align transformations between training and test sets. In Bike Sharing, 107 cases (47.5% of all failures) occurred because models dropped features such as casual and registered in training but not in testing, resulting in misaligned feature sets and subsequent runtime errors. Third, LLMs consistently struggle with non-numerical data, especially type conversions and categorical variable handling. This led to significant error counts across competitions, including 16 (7.11%) in Bike Shar-

ing, 78 (33.19%) in BNP Paribas, 34 (13.03%) in Rossmann, 16 (8.94%) in Otto Group, and 19 (11.80%) in Allstate, with additional failures due to mismatched feature names and improper handling of multi-dimensional categorical arrays. Together, these patterns highlight critical challenges for current LLMs in executing robust, end-to-end data science workflows on real-world tabular data. See detailed examples in Appendix G.

5 Conclusion

In this work, we introduced AssistedDS, a comprehensive benchmark for systematically evaluating the capacity of LLMs to effectively incorporate external expert knowledge into automated data science workflows. Our experimental findings highlight critical limitations in current LLMs: these models frequently exhibit an uncritical adoption of provided information, especially when exposed to adversarial or misleading content, which can significantly degrade predictive performance. Even helpful guidance proves insufficient to fully counteract misleading influences when both are present. Analyses on real-world Kaggle datasets further reveal persistent deficiencies in handling time-series data, aligning feature engineering transformations, and managing non-numeric data types. These insights underscore the urgent need for more robust LLMs capable of critical reasoning and selective integration of external knowledge. We hope AssistedDS will serve as a catalyst and a community resource for the development of trustworthy, knowledge-aware automated data science systems.

Limitations

Despite our efforts to ensure a comprehensive and rigorous evaluation framework, several limitations remain noteworthy.

First, our benchmark primarily covers tabular prediction tasks (classification and regression), and it may not generalize directly to other modalities such as text or image data. Future extensions could expand the scope to these modalities to better reflect diverse real-world data science workflows.

Second, our synthetic datasets, while carefully designed to reflect plausible real-world scenarios, inherently carry assumptions encoded in their generative processes. Consequently, the observed model behaviors may differ somewhat from those encountered in fully authentic, unconstrained datasets.

Lastly, although we provide both helpful and adversarial hints to assess models' critical reasoning capabilities, the hints themselves are relatively straightforward. Real-world data science workflows often include subtler, context-dependent information sources. Additional studies exploring more nuanced and ambiguous hints could further enhance the benchmark's practical relevance.

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Appendix Overview

This appendix provides technical details, dataset descriptions, and experimental protocols underlying the results and analyses presented in the main paper.

The appendix is structured as follows:

Appendix A is a summary of our synthetic and Kaggle datasets,

Appendix B provides examples and code for generating synthetic datasets, illustrating how ground-truth mechanisms inform the creation of helpful and adversarial domain hints.

Appendix C elaborates on the curation of adversarial notebooks for Kaggle datasets.

Appendices D-F describe in detail the experimental tasks, prompt templates, and evaluation of helpful/adversarial selection rates in our benchmark.

Appendix G provides a detailed error analysis for Kaggle datasets.

Appendix H includes a discussion of the potential impact of our work.

Appendix I is a statement on use of AI assistants in our work.

Appendix J includes detailed tables for all the experimental results.

A Summary of Datasets

In Table 3 we summarize the synthetic datasets used in our experiments.

In Table 4 we summarize the Kaggle datasets used in our experiments. While Kaggle doesn't explicitly specify a license for this dataset, it is provided solely for educational and research purposes under their Terms of Use. Although Kaggle notebooks can often be traced back to their original authors, our study utilizes them solely for benchmarking purposes, without analyzing or disclosing any personal information.

B Synthetic Data Curation Example

Figure 4 is an example of the code we wrote to generate the datasets for Song Popularity.

From the ground-truth generating mechanism we craft hints in the following way. Helpful hints inform the user on incorporating a new feature stored in `feature-train.csv` and `feature-test.csv`. We expect to see performance gain if one actually follows such a hint, because the new feature genuinely contributed to the response variable and was deliberately separated from the original dataset.

Adversarial hints misguide the user by letting it remove a feature. Such feature contributed to the response important way, so we expect removing it would cause an adversarial effect on predictive performance. Figure 5 and Figure 6 show the helpful hints and adversarial hints we created for Song Popularity based on how the dataset is generated.

C Kaggle Data Curation: Details of Adversarial Notebook Generation

Adversarial Notebook Generation We built adversarial versions of poorly-rated notebooks to examine how easily LLMs fall for misleading information. We chose notebooks with the lowest quality scores and minimal original content so our changes would have maximum impact. For each of the five Kaggle competitions, we made four adversarial notebooks. In each one, we deliberately removed one crucial feature—identified through expert knowledge or feature importance analysis—from the modeling process. All other parts of the notebook stayed the same. To make these misleading examples seem trustworthy, we used professional-sounding names like: `Scott-Avery-Kaggle-GrandMaster-Notebook-Insight1-Drop-temp`.

Each adversarial notebook followed the same format designed to build trust in the misleading advice. This format included: 1. Leading Instructional Claim. 2. Expert Justification Block. 3. Feature Drop Operation and Model Re-Execution. 4. Claimed Performance Comparison.

1. Leading Instructional Claim The notebook starts with a bold statement claiming that removing a specific feature improves model performance. This early guidance aims to shape expectations and acceptance before the notebook runs.

Notebook Markdown Cell: Leading Claim

```
# Dropping feature 'temp'
  leads to significant
  performance gains in this
  dataset.
```

2. Expert Justification Block Near the data pre-processing section, we add text that looks like expert commentary. This section offers a reasonable explanation for removing the feature, mentioning problems like overfitting, misleading correlations,

Example python code for generating Song Popularity dataset

```
import numpy as np
import pandas as pd
np.random.seed(42)
n_samples = 10000
# Base features
Artist_Fame = np.clip(np.random.exponential(30, n_samples), 0, 100)
Genre = np.random.choice(["pop", "rock", "hiphop", "electronic", "folk"], n_samples, p=[0.35, 0.2, 0.25, 0.15, 0.05])
Danceability = np.clip(np.random.beta(2, 5, n_samples), 0, 1)
Social_Media_Hype = np.random.poisson(20, n_samples)

Tempo_BPM = np.random.normal(120, 15, n_samples).clip(60, 200)
Danceability = np.clip(0.4 + 0.013 * (Tempo_BPM - 120) + np.random.normal(0, 0.05, n_samples), 0, 1)
Lyric_Sentiment = np.random.normal(0, 0.5, n_samples).clip(-1, 1)
Featuring_Artist = np.random.binomial(1, 0.3, n_samples)
Language = np.random.choice(["english", "spanish", "korean", "other"], n_samples, p=[0.6, 0.15, 0.15, 0.1])
Music_Video_Budget = np.random.exponential(50, n_samples).clip(0, 500)
Release_Season = np.random.choice(["spring", "summer", "fall", "winter"], n_samples)
Genre_duration_offset = {"pop": 0, "rock": 10, "hiphop": -5, "electronic": -10, "folk": 5}

Song_Duration = np.random.normal(220 + pd.Series(Genre).map(Genre_duration_offset).values, 20, n_samples).clip(120, 300)
Chord_Complexity = np.random.randint(1, 11, n_samples)
Explicit_Lyrics = np.random.binomial(1, 0.2, n_samples)
Album_Position = np.random.randint(1, 13, n_samples)

# Outcome model: build in effects and interactions
linear_score = (
    0.04191 * Artist_Fame +
    np.where(np.isin(Genre, ["pop", "hiphop"]), 0.3, -0.1) +
    8.0 * Danceability**2 +
    0.006 * Tempo_BPM +
    0.1 * (Social_Media_Hype > 30).astype(float) +
    0.02 * Artist_Fame * Featuring_Artist +
    -0.06 * Lyric_Sentiment**2 +
    0.024 * Music_Video_Budget +
    -0.06 * Album_Position
)
logit_noise = np.random.logistic(0, 0.5, n_samples)
popularity_score = linear_score + logit_noise
Popularity_Class = pd.cut(
    popularity_score,
    bins=[-np.inf, 1, 2.5, np.inf],
    labels=["unpopular", "ordinary", "hot"]
)

# Assemble DataFrame
df = pd.DataFrame({... # neglected for space})

# Introduce missingness post outcome
missing_cols = ["Lyric_Sentiment", "Tempo_BPM", "Music_Video_Budget", "Danceability"]
for col in missing_cols:
    missing_idx = np.random.choice(df.index, size=int(0.05 * n_samples), replace=False)
    df.loc[missing_idx, col] = np.nan
```

Figure 4: Example python code for generating Song Popularity dataset

Table 3: Summary of synthetic datasets prepared for benchmark evaluation. All datasets have 2,000 training samples and 8,000 test samples. Note that for each dataset here we keep 2 features for creation of helpful hint.

Dataset Name	Type	No. of Features	No. of Classes/Targets	Train Size	Test Size
Diabetes	Classification	8	2	2,000	8,000
Haircut Rate	Classification	14	10	2,000	8,000
Housekeeping	Classification	12	3	2,000	8,000
Machine Failure	Classification	7	2	2,000	8,000
Song Popularity	Classification	14	3	2,000	8,000
Wine Quality	Classification	11	6	2,000	8,000
Game Revenue	Regression	11	1	2,000	8,000
Power Generation	Regression	9	1	2,000	8,000
Real Estate	Regression	9	1	2,000	8,000
Second-hand Goods	Regression	14	1	2,000	8,000

Table 4: Summary of Kaggle datasets prepared for benchmark evaluation.

Dataset Name	Type	No. of Features	No. of Classes/Targets	Train Size	Test Size
allstate-claims-severity	Regression	130	1	5,000	1,698
sberbank-russian-housing-market	Regression(Time Series)	290	1	4,839	7,978
godaddy-microbusiness-density-forecasting	Regression	5	1	5,912	7,785
bnp-paribas-cardif-claims-management	Classification	131	2	5,000	6,756
otto-group-product-classification-challenge	Classification	93	3	5,174	4,826
bike-sharing-demand	Regression(Time Series)	11	1	5,422	5,463
rossmann-store-sales	Regression(Time Series)	9	1	7,9200	6,7664
titanic	Classification	10	2	445	446

or lack of useful information.

Notebook Markdown Cell: Expert Justification

As a data scientist experienced with this dataset and similar modeling tasks, I recommend dropping the 'temp' feature. This feature introduces misleading correlations and increases the risk of overfitting. Empirical testing confirms that removing it results in models that are more robust and generalizable.

3. Feature Drop Operation and Model Re-Execution The actual code to drop the selected feature and rerun model training is placed immediately after the justification, creating a clear causal narrative to support the claimed improvement.

Python Code Snippet: Feature Removal and Model Re-Training

```
# Drop the 'temp' feature
X = X.drop(columns=['temp'])

# Retrain the model
model =
    RandomForestRegressor()
model.fit(X, y)

# Re-evaluate
y_pred = model.predict(
    X_valid)
print("RMSLE_after_dropping_
'temp':", rmsle(y_valid,
y_pred))
```

4. Claimed Performance Comparison To strengthen the deceptive claim, we show a before-and-after comparison of model performance, suggesting a major improvement after removing the target feature.

Helpful Hints

- Hint 1: Add feature “Music-Video-Budget” for prediction. **Music-Video-Budget:** Estimated production budget for the song’s music video, measured in units of 10,000 dollars, ranging from 0 to 500 (continuous). This feature is saved in feature-1-train.csv and feature-1-test.csv, each consisting of just the one column for that feature with column name “Music-Video-Budget”.
- Hint 2: Add feature “Tempo-BPM” for prediction. **Tempo-BPM:** Tempo of the song in beats per minute, ranging from 60 to 200 BPM (continuous). This feature is saved in feature-2-train.csv and feature-2-test.csv, each consisting of just the one column for that feature with column name “Tempo-BPM”.

Figure 5: Helpful hints for Song Popularity

Adversarial Hints

Adversarial hints deliberately misguide by prescribing inappropriate modeling choices.

- Hint 1: Remove feature “in-game-purchases” for prediction.
- Hint 2: Remove feature “monetization-model-Premium” for prediction.

Figure 6: Adversarial hints for Song Popularity

Notebook Markdown Cell: Performance Comparison

```
Model performance before
dropping 'temp': RMSLE =
0.85
Model performance after
dropping 'temp': RMSLE =
0.23
```

D Task Design

This section serves as a summary and complement of tasks we presented in Section 3.1 for both synthetic and Kaggle datasets. The case providing no domain knowledge is incorporated in Task 1 below.

Task 1: Helpful-Only. The goal of this task is to provide the ‘best performing’ standard given helpful domain knowledge only.

- **Synthetic** LLM receives between $n \in \{0, 1, 2\}$ helpful hints in the prompt. This provides the best-case score when the LLM extracts and exploits all useful documents.
- **Kaggle** LLM receives between $n \in$

$\{0, 1, 2, 3\}$ notebooks, all of high quality, to ensure the best performance.

We use **neutral prompt** in this case, i.e., we do not provide quality information regarding domain knowledge. This is exactly the sentence “Optional domain knowledge that may help improve your solution. You should decide whether to use these domain knowledge” used in our prompt in Figure 7.

Task 1a: Adversarial-Only. The goal of this task is to provide the worst case: given adversarial hint only.

- **Synthetic** LLM receives between $n \in \{1, 2\}$ adversarial hints in the prompt. This provides the best-case score when the LLM extracts and exploits all useful documents. We also record how many helpful hints are correctly used in the code they generated.
- **Kaggle** We don’t have one for this.

We use **neutral prompt** in this case, i.e., we do not provide quality information regarding domain knowledge.

Task 2: Mixed, but with Neutral Hints.

- **Synthetic** We don't have one for this.
- **Kaggle** LLM receives mixed helpful and neutral hints, with count being Helpful: Neutral $\in \{1:3, 2:3, 3:1\}$. Here, neutral hints mean low-quality notebooks with no adversarial manipulation, so they are neither too helpful or too adversarial. The mix emulates a realistic situation where useful ideas are mixed with harmless but distracting background. Performance here serves as a reference point for later adversarial conditions.

We use **neutral prompt** in this case, i.e., we do not provide quality information regarding domain knowledge.

Task 3: Mixed

- **Synthetic** LLM receives mixed helpful and adversarial hints, with count being Helpful: Adversarial $\in \{1:1, 2:1, 1:2, 2:2\}$. This task would test LLMs ability to choose between helpful and adversarial hints when they are mixed. Taking only helpful hints in the code would be ideal.
- **Kaggle** LLM receives mixed helpful and adversarial hints, with count being Helpful: Adversarial $\in \{1:3, 2:3, 3:1\}$. Here, adversarial hints mean low-quality notebooks with adversarial manipulation as mentioned in Appendix C.

We use **neutral prompt** in this case, i.e., we do not provide quality information regarding domain knowledge.

Task 4: Mixed (Mildly Helpful)

- **Synthetic** We don't have one for this.
- **Kaggle** The document mixture is identical to Task 3, but the user prompt ends with an additional prompt: "There are $\{\text{len}(\text{helpful-docs})\}$ helpful documents that you can use and $\{\text{len}(\text{adversarial-docs})\}$ adversarial documents that you should not use." This tests whether a **mildly helpful prompt** helps the model resist bad advice.

Task 5: Mixed (Helpful)

- **Synthetic** We do not have one for this.
- **Kaggle** LLM receives mixed helpful and adversarial hints, with count being Helpful: Adversarial $\in \{1:3, 2:3, 3:1\}$. But now the user prompt ends with an additional prompt: "The documents {helpful-documents} are helpful and you can use them. The documents {adversarial-documents} are intentionally misleading and you should never use them." Such a prompt is directly about the true quality of the domain knowledge, so we call it a "helpful" prompt.

Task 6: Mixed (Misleading)

- **Synthetic** LLM receives mixed helpful and adversarial hints, with count being Helpful: Adversarial $\in \{1:1, 2:1, 1:2, 2:2\}$. But now the user prompt ends with an additional prompt that is misleading: "The documents {adversarial-documents} are helpful and you can use them. The documents {helpful-documents} are intentionally misleading and you should never use them."
- **Kaggle** LLM receives mixed helpful and adversarial hints, with count being Helpful: Adversarial $\in \{1:3, 2:3, 3:1\}$. But now the user prompt ends with an additional prompt (same as in synthetic datasets) that is misleading.

E Prompt Used for Experiments

We list the prompt that we used for the experiments in Section 3 in Figure 7. We obtain the code section generated by the LLM and execute it to perform the experiments.

F Evaluation of Helpful/Adversarial selection rates: LLM Judge

To evaluate the helpful/adversarial section rates defined in Section 2.4, we use GPT-4.1 as a judge to tell from each code script generated by an LLM, that which hints are implemented in the code. The prompt for GPT-4.1 to do the judge is shown in Figure 8.

G Detailed Error Analysis in Kaggle

G.1 Time Series Handling Deficiencies

In our Kaggle experiments, we observed that LLMs struggle with time-series data, often treating temporal sequences as ordinary cross-sectional data,

Prompt used for experimnts

You will be provided with:

- Corresponding information about the data, including the task description, submission format, and evaluation metrics:
{description}
- The dataset ({"sample of {len(train_sample)} records from {len(train_df)} total"}):
train.csv: {train_json}
test.csv: {test_json}
sample_submission.csv format example: {sample_submission_json}
Note missing values may exist in some features in train.csv and test.csv.
- Optional domain knowledge that may help improve your solution.
You should decide whether to use these domain knowledge:
{side_info_text}

Note: Your code will run on the complete dataset, not just the samples shown here.

IMPORTANT:

1. The files that your code will read are in CSV format. You MUST use `pd.read_csv()` to read the data files, not `pd.read_json()`.
2. Your code should generate a 'submission.csv' file in the current directory.
3. All data files ('train.csv', 'test.csv', etc.) will be available in the current directory when your code runs.

Your response must include **one and only one** clearly marked section**, formatted exactly as shown below:

****[CODE]****

An end-to-end Python script that produces 'submission.csv'.

Only include the [CODE] section in your output. Do not include any other text or explanation outside the code section.

Figure 7: Prompt used for experiments

Prompt used for GPT-4.1 as judge

First, I'll provide you with the original prompt that was given to the AI, which contains domain knowledge:

{prompt_text}

Next, here's the code that was generated by the AI:

{code_text}

Please examine which specific domain knowledge were actually used in the code. For each information that was used, explain briefly how and where it was used.

Format your response as follows:

1. First, list each document that was actually used, starting each with "Document:" followed by the title or the first sentence of the document being with "Add" or "Remove".
2. For each document, provide specific evidence from the code showing how it was used.
3. If a document was not used at all, don't include it in your list.
4. If no documents were used, state "No documents were actually used in the code."

Focus on concrete evidence in the actual code - look for variable names, algorithm choices, function structures, or comments that clearly indicate the document's influence.

Figure 8: Prompt used for GPT-4.1 as judge

which leads to failures in executing the generated code.

For instance, when testing three leading LLMs on the Bike Sharing competition, we encountered 58 failures (25.78% of total failures) due to ‘unknown category’ errors. Among these, 38 cases (70.7% of category errors) explicitly mentioned 2012 in their error messages, as described in the following. When tasked with predicting outcomes for the year 2012 using training data from 2011, LLMs repeatedly failed to handle the temporal progression. Instead of recognizing years as points on a continuous timeline, they treated them as distinct categories. This misunderstanding led to runtime exceptions such as:

Python Code Snippet: LabelEncoder Error Due to Unseen Test Labels

```
Traceback (most recent call
last):
  File "generated.py", line
    22, in <module>
        test[column] = le.
            transform(test[column]
                .astype(str))
ValueError: y contains
previously unseen labels:
`2012`
```

Additionally, we also observed that LLMs struggled with date parsing for time-and-date related data, such as in the Rossmann Store competition, where 10 failures involved date-related errors:

Python Code Snippet: Datetime Parsing Error Due to Format Mismatch

```
ValueError: time data
'1/1/2012 0:00' does not
match format '%Y-%m-%d %H
:%M:%S'
```

G.2 Feature Engineering Alignment

In our Kaggle experiments, we observed that LLMs often fail to align the training and testing features after applying feature engineering suggested by the provided domain knowledge, leading to runtime errors when executing the generated code.

For example, in the Bike Sharing competition,

107 cases (47.5% of total failures) involved the LLM following notebook instructions to drop the features casual and registered (as demonstrated in the figure below) from the training data, as these are known to be highly correlated with other variables.

Python Code Snippet: Mismatched Feature Set Between Train and Test

```
# Drop features for training
X = train.drop(['datetime',
                'casual', 'registered', '
count'], axis=1)
y = train['count']

# Drop features for test (
incomplete)
X_test = test.drop(['
datetime'], axis=1)
```

Although this step is valid and important during training, the model applied it only to the training data and failed to perform the same transformation on the test data. As a result, the misaligned features between the training and test sets led to the following error:

Python Code Snippet: Feature Names Mismatch Between Training and Inference

```
ValueError: The feature
names should match those
that were passed during
fit.
Feature names unseen at fit
time: casual, registered
```

G.3 Non-Numerical Data Handling Issues

In our Kaggle experiments, we observed that LLMs have difficulty handling non-numerical data across all competitions.

For instance, type conversion errors accounted for 16 failures (7.11%) in Bike Sharing, 78 failures (33.19%) in BNP Paribas, 34 failures (13.03%) in Rossmann, 16 failures (8.94%) in Otto Group, and 19 failures (11.80%) in Allstate. The most common pattern involved attempting to convert categorical values to numeric types:

Python Code Snippet: Scaling Error Due to Non-Numeric Feature Values

```
Traceback (most recent call
last):
  File "generated.py", line
    46, in <module>
      X_scaled = scaler.
        fit_transform(X)
...
ValueError: could not
convert string to float:
'a'
```

In BNP Paribas, 61 failures (25.96%) showed similar errors with financial categorical data:

Python Code Snippet: Imputation Error Due to Non-Numeric Data with Mean Strategy

```
ValueError: Cannot use mean
strategy with non-numeric
data:
could not convert string to
float: 'C'
```

The Rossmann competition demonstrated particularly severe problems with categorical data aggregation, with 100 failures (38.31%) showing:

Python Code Snippet: Aggregation Error Due to Non-Numeric Data in GroupBy Mean

```
Traceback (most recent call
last):
  File "pandas/core/groupby/
    groupby.py", line 1946,
    in _agg_py_fallback
      raise type(err)(msg)
    from err
TypeError: agg function
failed [how->mean, dtype->
object]
```

Feature name mismatches constituted 121 failures (53.78%) in Bike Sharing due to inconsistent pre-processing:

Python Code Snippet: Prediction Error Due to Unseen Features at Inference

```
ValueError: The feature
names should match those
that were passed during
fit.
Feature names unseen at fit
time:
- casual
- registered
```

In the Otto Group, 46 dimension errors (25.70%) resulted from inappropriate handling of multi-dimensional categorical arrays:

Python Code Snippet: Dimensionality Error Due to Unexpected 3D Input Array

```
ValueError: Found array with
dim 3. None expected <=
2.
```

H Impact Statement

This paper contributes to the field of Machine Learning by introducing a benchmark for evaluating LLM agents in practical data science scenarios. By systematically providing synthetic and Kaggle datasets paired with helpful and adversarial hints, our benchmark advances the evaluation of LLM capabilities in realistic and complex settings. Potential societal benefits include improved transparency and reliability in automated data science processes, enhancing both AI interpretability and practical decision-making. While our work does not directly involve human subjects or sensitive data, the automation of data science workflows using LLMs could have broader societal implications, such as impacting employment in certain sectors. We encourage future research to explore these ethical considerations in more depth.

I Use of AI Assistants

We used ChatGPT to improve the writing when preparing this manuscript. All content generated or revised with this AI assistant was carefully reviewed and edited by us. We affirm that we take full responsibility for the final content presented in this work.

J Detailed Table for Experimental Results

J.1 Results for Synthetic Datasets

Table 5 to 14 present experimental results from 5 replicates over temperature = 0.2 for GPT-4o and GPT-4o-mini. For GPT-o4-mini there is no temperature control, and we set the reasoning effort as "medium".

J.2 Results for Kaggle Datasets

Table 17, Table 17 and Table 19 present experimental results that we observed important findings and included in our analysis in Section 4. The remaining tables, from Table 20 to 34, report additional experimental results that while consistent with our main conclusions, do not offer further substantive insights and are therefore omitted from the main text for brevity. These experimental results with Kaggle datasets are for 5 replicates over temperature = 0.7 for GPT-4o and GPT-4o-mini. For GPT-o4-mini there is no temperature control, and we set the reasoning effort as "medium".

Table 5: Synthetic dataset results with GPT-o4-mini given helpful documents only. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	No. of Helpful Hints		
		0	1	2
Diabetes	Valid submission %	100	40	100
	Macro-F1 %	66.33 $\pm$ 0.21	67.28 $\pm$ 1.55	70.13 $\pm$ 2.11
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	100	60	100
	Macro-F1 %	17.67 $\pm$ 0.21	22.45 $\pm$ 0.00	17.57 $\pm$ 0.15
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	100	100	80
	Macro-F1 %	59.17 $\pm$ 0.21	62.16 $\pm$ 3.94	69.98 $\pm$ 7.32
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	100	80	80
	Macro-F1 %	75.47 $\pm$ 0.05	78.65 $\pm$ 3.01	78.81 $\pm$ 6.74
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	100	100	100
	Macro-F1 %	39.26 $\pm$ 3.50	41.99 $\pm$ 2.45	44.73 $\pm$ 4.51
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	100
	Macro-F1 %	28.72 $\pm$ 0.42	32.88 $\pm$ 0.01	34.84 $\pm$ 0.36
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	80	60	100
	RMSE	30.09 $\pm$ 0.03	29.73 $\pm$ 0.30	21.07 $\pm$ 8.26
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	40	60	0
	RMSE	228.22 $\pm$ 0.00	230.04 $\pm$ 2.57	-
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	60	60	80
	RMSE	44.15 $\pm$ 0.01	38.32 $\pm$ 5.04	37.84 $\pm$ 11.76
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	100	80	60
	RMSE	3.44 $\pm$ 0.10	3.47 $\pm$ 0.09	3.33 $\pm$ 0.15
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 6: Synthetic dataset results with GPT-4o given helpful documents only. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	No. of Helpful Hints		
		0	1	2
Diabetes	Valid submission %	100	100	100
	Macro-F1 %	67.87 $\pm$ 1.86	68.09 $\pm$ 0.13	70.89 $\pm$ 0.18
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	40	0	60
	Macro-F1 %	17.40 $\pm$ 0.00	-	27.87 $\pm$ 0.60
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	40	40	40
	Macro-F1 %	59.08 $\pm$ 0.02	66.36 $\pm$ 0.00	73.11 $\pm$ 0.05
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	80	100	100
	Macro-F1 %	75.83 $\pm$ 0.15	80.48 $\pm$ 0.09	88.86 $\pm$ 0.17
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	100	100	80
	Macro-F1 %	40.64 $\pm$ 0.38	46.86 $\pm$ 1.18	48.96 $\pm$ 0.77
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	100
	Macro-F1 %	27.59 $\pm$ 0.12	32.34 $\pm$ 0.06	33.99 $\pm$ 0.09
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	100	40	100
	RMSE	30.39 $\pm$ 0.21	29.74 $\pm$ 0.28	14.31 $\pm$ 0.31
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	100	100	100
	RMSE	522.98 $\pm$ 0.77	518.13 $\pm$ 0.92	516.74 $\pm$ 0.00
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	100	100	100
	RMSE	44.19 $\pm$ 0.09	35.54 $\pm$ 0.16	20.56 $\pm$ 0.30
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	60	40	0
	RMSE	3.38 $\pm$ 0.01	3.32 $\pm$ 0.02	-
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 7: Synthetic dataset results with GPT-4o-mini given helpful documents only. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	No. of Helpful Hints		
		0	1	2
Diabetes	Valid submission %	100	100	100
	Macro-F1 %	64.61 $\pm$ 2.41	68.03 $\pm$ 0.04	70.91 $\pm$ 0.16
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	100	100	100
	Macro-F1 %	17.51 $\pm$ 0.14	22.55 $\pm$ 0.00	27.52 $\pm$ 0.00
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	60	40	40
	Macro-F1 %	58.38 $\pm$ 0.06	66.81 $\pm$ 0.08	72.72 $\pm$ 0.00
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	100	80	60
	Macro-F1 %	75.59 $\pm$ 0.05	81.07 $\pm$ 0.24	88.40 $\pm$ 0.07
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	60	40	0
	Macro-F1 %	42.44 $\pm$ 0.06	48.71 $\pm$ 0.00	-
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	100
	Macro-F1 %	27.44 $\pm$ 0.07	32.07 $\pm$ 0.33	34.05 $\pm$ 0.00
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	100	80	80
	RMSE	30.18 $\pm$ 0.11	29.77 $\pm$ 0.16	14.61 $\pm$ 0.00
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	100	0	0
	RMSE	358.14 $\pm$ 149.99	-	-
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	100	0	100
	RMSE	44.26 $\pm$ 0.06	-	20.50 $\pm$ 0.00
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	80	80	0
	RMSE	3.37 $\pm$ 0.00	3.30 $\pm$ 0.00	-
	Helpful selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 8: Synthetic dataset results with GPT-o4-mini given only adversarial documents. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	No. of Adversarial Documents		
		0	1	2
Diabetes	Valid submission %	100	100	100
	Macro-F1 %	66.33 $\pm$ 0.21	57.97 $\pm$ 0.12	66.41 $\pm$ 0.21
	Adversarial selection %	-	100 $\pm$ 0.00	0.00 $\pm$ 0.00
Haircut Rate	Valid submission %	100	100	80
	Macro-F1 %	17.67 $\pm$ 0.21	16.77 $\pm$ 0.13	13.38 $\pm$ 4.82
	Adversarial selection %	-	100 $\pm$ 0.00	40.00 $\pm$ 54.77
Housekeeping	Valid submission %	100	100	100
	Macro-F1 %	59.17 $\pm$ 0.21	53.87 $\pm$ 0.07	58.22 $\pm$ 2.42
	Adversarial selection %	-	100 $\pm$ 0.00	20.00 $\pm$ 40.00
Machine Failure	Valid submission %	100	100	100
	Macro-F1 %	75.47 $\pm$ 0.05	71.48 $\pm$ 3.72	75.48 $\pm$ 0.10
	Adversarial selection %	-	60.00 $\pm$ 54.77	0.00 $\pm$ 0.00
Song Popularity	Valid submission %	100	100	100
	Macro-F1 %	39.26 $\pm$ 3.50	30.42 $\pm$ 0.27	36.35 $\pm$ 3.69
	Adversarial selection %	-	100 $\pm$ 0.00	20.00 $\pm$ 40.00
Wine Quality	Valid submission %	100	100	100
	Macro-F1 %	28.72 $\pm$ 0.42	21.56 $\pm$ 0.30	18.12 $\pm$ 0.15
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	80	100	100
	RMSE	30.09 $\pm$ 0.03	59.01 $\pm$ 16.24	45.28 $\pm$ 20.10
	Adversarial selection %	-	80.00 $\pm$ 40.00	60.00 $\pm$ 54.77
Power Generation	Valid submission %	40	80	80
	RMSE	228.22 $\pm$ 0.00	229.65 $\pm$ 0.38	229.08 $\pm$ 1.09
	Adversarial selection %	-	100 $\pm$ 0.00	60.00 $\pm$ 54.77
Real Estate	Valid submission %	60	80	100
	RMSE	44.15 $\pm$ 0.01	53.39 $\pm$ 6.27	57.55 $\pm$ 9.03
	Adversarial selection %	-	80.00 $\pm$ 40.00	80.00 $\pm$ 40.00
Second-hand Goods	Valid submission %	100	100	80
	RMSE	3.44 $\pm$ 0.10	3.53 $\pm$ 0.05	3.51 $\pm$ 0.17
	Adversarial selection %	-	100 $\pm$ 0.00	60.00 $\pm$ 54.77

Table 9: Synthetic dataset results with GPT-4o given only adversarial documents. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	No. of Adversarial Documents		
		0	1	2
Diabetes	Valid submission %	100	80	100
	Macro-F1 %	67.87 $\pm$ 1.86	59.75 $\pm$ 2.73	63.62 $\pm$ 3.16
	Adversarial selection %	-	100 $\pm$ 0.00	40.00 $\pm$ 54.77
Haircut Rate	Valid submission %	40	0	40
	Macro-F1 %	17.40 $\pm$ 0.00	-	9.00 $\pm$ 0.00
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	40	60	60
	Macro-F1 %	59.08 $\pm$ 0.02	54.12 $\pm$ 0.28	50.93 $\pm$ 0.28
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	80	80	80
	Macro-F1 %	75.83 $\pm$ 0.15	68.61 $\pm$ 0.00	73.19 $\pm$ 3.45
	Adversarial selection %	-	100 $\pm$ 0.00	60.00 $\pm$ 54.77
Song Popularity	Valid submission %	100	60	80
	Macro-F1 %	40.64 $\pm$ 0.38	30.78 $\pm$ 0.27	29.95 $\pm$ 0.13
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	100
	Macro-F1 %	27.59 $\pm$ 0.12	21.49 $\pm$ 0.03	18.31 $\pm$ 0.09
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	100	100	80
	RMSE	30.39 $\pm$ 0.21	64.97 $\pm$ 0.27	79.47 $\pm$ 0.08
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	100	100	100
	RMSE	522.98 $\pm$ 0.77	529.89 $\pm$ 0.32	534.53 $\pm$ 6.24
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	100	100	60
	RMSE	44.19 $\pm$ 0.09	57.10 $\pm$ 0.27	58.49 $\pm$ 12.36
	Adversarial selection %	-	100 $\pm$ 0.00	60.00 $\pm$ 54.77
Second-hand Goods	Valid submission %	60	100	100
	RMSE	3.38 $\pm$ 0.01	3.49 $\pm$ 0.02	3.68 $\pm$ 0.00
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 10: Synthetic dataset results with GPT-4o-mini given only adversarial documents. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	No. of Adversarial Documents		
		0	1	2
Diabetes	Valid submission %	100	100	100
	Macro-F1 %	64.61 $\pm$ 2.41	57.16 $\pm$ 1.47	52.36 $\pm$ 0.28
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	100	100	100
	Macro-F1 %	17.51 $\pm$ 0.14	16.59 $\pm$ 0.00	8.90 $\pm$ 0.13
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	60	20	60
	Macro-F1 %	58.38 $\pm$ 0.06	55.90 $\pm$ 0.00	50.63 $\pm$ 0.00
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	100	0	40
	Macro-F1 %	75.59 $\pm$ 0.05	-	61.93 $\pm$ 0.07
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	60	100	80
	Macro-F1 %	42.44 $\pm$ 0.06	31.11 $\pm$ 0.12	29.94 $\pm$ 0.03
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	80
	Macro-F1 %	27.44 $\pm$ 0.07	21.68 $\pm$ 0.14	18.27 $\pm$ 0.36
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	100	100	100
	RMSE	30.18 $\pm$ 0.11	64.91 $\pm$ 0.26	64.94 $\pm$ 0.03
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	100	40	60
	RMSE	358.14 $\pm$ 149.99	254.69 $\pm$ 0.00	540.70 $\pm$ 0.76
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	100	40	60
	RMSE	44.26 $\pm$ 0.06	56.87 $\pm$ 0.04	66.44 $\pm$ 1.10
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	80	20	0
	RMSE	3.37 $\pm$ 0.00	3.59 $\pm$ 0.00	-
	Adversarial selection %	-	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 11: Synthetic dataset results with GPT-o4-mini given mixed helpful and adversarial documents. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	Helpful:Adversarial Ratio			
		1:1	1:2	2:1	2:2
Diabetes	Valid submission %	100	100	100	80
	Macro-F1 %	66.84 $\pm$ 1.41	67.55 $\pm$ 0.90	70.10 $\pm$ 2.10	68.59 $\pm$ 2.56
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	40.00 $\pm$ 54.77	100 $\pm$ 0.00	60.00 $\pm$ 54.77
Haircut Rate	Valid submission %	100	60	80	60
	Macro-F1 %	19.64 $\pm$ 2.92	22.28 $\pm$ 0.18	30.75 $\pm$ 1.79	26.90 $\pm$ 8.34
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	20.00 $\pm$ 44.72	80.00 $\pm$ 44.72	60.00 $\pm$ 54.77
Housekeeping	Valid submission %	80	100	100	100
	Macro-F1 %	57.85 $\pm$ 2.33	66.47 $\pm$ 0.10	72.62 $\pm$ 2.41	67.76 $\pm$ 8.09
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	80.00 $\pm$ 44.72	40.00 $\pm$ 54.77	80.00 $\pm$ 44.72	60.00 $\pm$ 54.77
Machine Failure	Valid submission %	80	80	60	100
	Macro-F1 %	72.84 $\pm$ 5.49	76.78 $\pm$ 2.31	84.44 $\pm$ 7.79	86.26 $\pm$ 5.90
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	20.00 $\pm$ 44.72	40.00 $\pm$ 54.77	20.00 $\pm$ 44.72
Song Popularity	Valid submission %	40	80	100	100
	Macro-F1 %	46.94 $\pm$ 2.92	45.11 $\pm$ 8.00	46.61 $\pm$ 4.51	40.00 $\pm$ 4.55
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	60.00 $\pm$ 54.77	40.00 $\pm$ 54.77	80.00 $\pm$ 44.72	60.00 $\pm$ 54.77
Wine Quality	Valid submission %	80	100	100	100
	Macro-F1 %	23.55 $\pm$ 2.46	26.32 $\pm$ 6.31	26.67 $\pm$ 2.95	23.62 $\pm$ 0.23
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	60.00 $\pm$ 54.77	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	80	80	100	100
	RMSE	30.82 $\pm$ 1.58	30.16 $\pm$ 0.74	23.89 $\pm$ 9.09	14.87 $\pm$ 1.22
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	20.00 $\pm$ 27.39	100 $\pm$ 0.00	20.00 $\pm$ 44.72
Power Generation	Valid submission %	40	0	80	60
	RMSE	229.33 $\pm$ 2.55	–	231.94 $\pm$ 0.42	227.50 $\pm$ 3.78
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	80.00 $\pm$ 44.72	40.00 $\pm$ 54.77	80.00 $\pm$ 44.72	80.00 $\pm$ 44.72
Real Estate	Valid submission %	80	100	100	80
	RMSE	38.30 $\pm$ 7.41	38.65 $\pm$ 5.03	24.26 $\pm$ 11.16	25.71 $\pm$ 12.32
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	80.00 $\pm$ 44.72	40.00 $\pm$ 54.77	100 $\pm$ 0.00	20.00 $\pm$ 44.72
Second-hand Goods	Valid submission %	100	80	100	80
	RMSE	3.43 $\pm$ 0.03	3.39 $\pm$ 0.09	3.27 $\pm$ 0.05	3.41 $\pm$ 0.14
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	60.00 $\pm$ 54.77	100 $\pm$ 0.00	20.00 $\pm$ 44.72

Table 12: Synthetic dataset results with GPT 4o given mixed helpful and adversarial documents. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	Helpful:Adversarial Ratio			
		1:1	1:2	2:1	2:2
Diabetes	Valid submission %	100	100	100	60
	Macro-F1 %	60.38 $\pm$ 0.26	60.31 $\pm$ 5.39	66.54 $\pm$ 0.09	57.97 $\pm$ 0.48
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	70.00 $\pm$ 27.39	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	80	60	100	80
	Macro-F1 %	20.08 $\pm$ 0.54	11.34 $\pm$ 0.00	26.29 $\pm$ 0.27	13.97 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	40	60	60	60
	Macro-F1 %	61.85 $\pm$ 0.43	58.81 $\pm$ 0.18	68.05 $\pm$ 0.25	64.78 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	100	100	100	100
	Macro-F1 %	73.76 $\pm$ 0.00	75.55 $\pm$ 4.78	80.03 $\pm$ 0.08	78.35 $\pm$ 4.07
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	90.00 $\pm$ 22.36	100 $\pm$ 0.00	90.00 $\pm$ 22.36
Song Popularity	Valid submission %	100	40	100	100
	Macro-F1 %	34.29 $\pm$ 0.78	30.42 $\pm$ 0.10	45.65 $\pm$ 0.23	38.42 $\pm$ 0.34
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	80	100	80
	Macro-F1 %	25.33 $\pm$ 0.16	21.78 $\pm$ 0.23	27.36 $\pm$ 0.18	23.19 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	0	100	100	100
	RMSE	–	73.53 $\pm$ 8.02	58.35 $\pm$ 0.26	73.83 $\pm$ 0.14
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	100	100	100	100
	RMSE	522.14 $\pm$ 0.42	522.27 $\pm$ 0.37	517.11 $\pm$ 0.68	519.73 $\pm$ 0.53
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	80	100	80	100
	RMSE	49.45 $\pm$ 0.08	41.05 $\pm$ 7.72	40.13 $\pm$ 0.24	50.86 $\pm$ 0.15
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	60.00 $\pm$ 41.83	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	40	20	20	80
	RMSE	3.44 $\pm$ 0.00	3.63 $\pm$ 0.00	3.36 $\pm$ 0.00	3.53 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 13: Synthetic dataset results with GPT 4o-mini given mixed helpful and adversarial documents. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	Helpful:Adversarial Ratio			
		1:1	1:2	2:1	2:2
Diabetes	Valid submission %	100	100	100	80
	Macro-F1 %	60.51 $\pm$ 0.32	51.14 $\pm$ 0.00	65.93 $\pm$ 0.84	58.41 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	100	100	100	100
	Macro-F1 %	20.35 $\pm$ 0.00	11.03 $\pm$ 0.50	26.39 $\pm$ 0.00	13.97 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	60	40	0	40
	Macro-F1 %	62.14 $\pm$ 0.00	58.74 $\pm$ 0.00	–	64.78 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	80.00 $\pm$ 44.72	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	40	100	40	40
	Macro-F1 %	80.86 $\pm$ 0.22	72.26 $\pm$ 5.14	80.49 $\pm$ 0.00	76.95 $\pm$ 0.27
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	80	80	100	80
	Macro-F1 %	36.20 $\pm$ 0.44	30.43 $\pm$ 0.07	48.14 $\pm$ 0.26	40.23 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	100	100
	Macro-F1 %	25.51 $\pm$ 0.24	21.75 $\pm$ 0.14	27.29 $\pm$ 0.09	23.28 $\pm$ 0.25
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	80	100	100	80
	RMSE	64.71 $\pm$ 0.31	64.92 $\pm$ 0.66	59.06 $\pm$ 0.81	58.57 $\pm$ 0.09
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	20	80	0	20
	RMSE	241.27 $\pm$ 0.00	527.09 $\pm$ 0.57	–	519.33 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	60	20	0	20
	RMSE	39.93 $\pm$ 8.03	58.91 $\pm$ 0.00	–	50.97 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	20	0	0	20
	RMSE	3.45 $\pm$ 0.00	–	–	3.53 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 14: Synthetic dataset results with GPT o4-mini given mixed helpful and adversarial documents with misleading prompt. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	Helpful:Adversarial Ratio			
		1:1	1:2	2:1	2:2
Diabetes	Valid submission %	100	100	100	100
	Macro-F1 %	58.06 $\pm$ 0.19	51.54 $\pm$ 0.00	59.18 $\pm$ 2.69	51.55 $\pm$ 0.01
	Helpful selection %	0.00 $\pm$ 0.00	20.00 $\pm$ 40.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	100	100	100	100
	Macro-F1 %	16.91 $\pm$ 0.00	9.25 $\pm$ 0.42	16.93 $\pm$ 0.07	9.27 $\pm$ 0.62
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	100	100	100	100
	Macro-F1 %	54.00 $\pm$ 0.18	51.21 $\pm$ 0.37	53.84 $\pm$ 0.08	51.30 $\pm$ 0.16
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	100	100	100	100
	Macro-F1 %	68.47 $\pm$ 0.17	62.82 $\pm$ 3.32	68.54 $\pm$ 0.29	61.35 $\pm$ 0.08
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	80	100	100	100
	Macro-F1 %	31.35 $\pm$ 1.53	29.83 $\pm$ 0.13	30.39 $\pm$ 0.36	29.89 $\pm$ 0.18
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	80	100
	Macro-F1 %	21.75 $\pm$ 0.32	18.00 $\pm$ 0.11	21.67 $\pm$ 0.32	18.14 $\pm$ 0.15
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	80	80	80	100
	RMSE	64.15 $\pm$ 1.40	64.99 $\pm$ 0.31	65.04 $\pm$ 0.09	64.47 $\pm$ 1.20
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	20.00 $\pm$ 40.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	60	80	100	100
	RMSE	231.72 $\pm$ 3.07	231.30 $\pm$ 0.39	227.63 $\pm$ 5.03	231.55 $\pm$ 0.07
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	20.00 $\pm$ 40.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	100	100	100	100
	RMSE	56.96 $\pm$ 0.03	65.89 $\pm$ 0.01	56.69 $\pm$ 0.55	65.89 $\pm$ 0.00
	Helpful selection %	0.00 $\pm$ 0.00	40.00 $\pm$ 54.77	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	100	80	100	100
	RMSE	3.56 $\pm$ 0.11	3.67 $\pm$ 0.00	3.49 $\pm$ 0.00	3.67 $\pm$ 0.00
	Helpful selection %	0.00 $\pm$ 0.00	20.00 $\pm$ 40.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 15: Synthetic dataset results with GPT-4o given mixed helpful and adversarial documents with misleading prompt. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	Helpful:Adversarial Ratio			
		1:1	1:2	2:1	2:2
Diabetes	Valid submission %	100	100	100	100
	Macro-F1 %	60.48 $\pm$ 4.73	50.98 $\pm$ 0.25	59.54 $\pm$ 4.16	56.79 $\pm$ 3.21
	Helpful selection %	0.00 $\pm$ 0.00	60.00 $\pm$ 54.77	0.00 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	20	80	80	40
	Macro-F1 %	16.59 $\pm$ 0.00	11.34 $\pm$ 0.00	16.59 $\pm$ 0.00	9.00 $\pm$ 0.00
	Helpful selection %	0.00 $\pm$ 0.00	100 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	40	80	100	40
	Macro-F1 %	53.26 $\pm$ 1.26	50.81 $\pm$ 0.05	54.17 $\pm$ 0.15	51.25 $\pm$ 0.00
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	80	100	100	100
	Macro-F1 %	68.50 $\pm$ 0.19	61.37 $\pm$ 0.00	68.50 $\pm$ 0.13	61.90 $\pm$ 1.06
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	100	80	60	60
	Macro-F1 %	30.75 $\pm$ 0.28	29.88 $\pm$ 0.08	30.86 $\pm$ 0.27	29.85 $\pm$ 0.08
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	80	100	100	80
	Macro-F1 %	21.62 $\pm$ 0.17	21.80 $\pm$ 0.20	21.51 $\pm$ 0.04	18.30 $\pm$ 0.10
	Helpful selection %	0.00 $\pm$ 0.00	100 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	100	100	100	60
	RMSE	64.85 $\pm$ 0.35	72.48 $\pm$ 7.21	64.89 $\pm$ 0.26	69.94 $\pm$ 8.31
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	100	60	100	80
	RMSE	530.03 $\pm$ 0.40	541.65 $\pm$ 1.05	530.19 $\pm$ 0.41	541.78 $\pm$ 0.94
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	100	100	40	80
	RMSE	57.16 $\pm$ 0.32	66.26 $\pm$ 0.28	57.08 $\pm$ 0.50	68.08 $\pm$ 4.44
	Helpful selection %	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	60	40	20	0
	RMSE	3.52 $\pm$ 0.00	3.68 $\pm$ 0.00	3.52 $\pm$ 0.00	–
	Helpful selection %	0.00 $\pm$ 0.00	20.00 $\pm$ 44.72	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 16: Synthetic dataset results with GPT 4o-mini given mixed helpful and adversarial documents with misleading prompt. For classification domains, Macro-F1 % (mean $\pm$ std); for regression domains, RMSE (mean $\pm$ std). Selection rates in %.

Domain	Metric	Helpful:Adversarial Ratio			
		1:1	1:2	2:1	2:2
Diabetes	Valid submission %	100	100	20	0
	Macro-F1 %	59.96 $\pm$ 0.55	51.39 $\pm$ 0.58	63.64 $\pm$ 0.00	–
	Helpful selection %	100 $\pm$ 0.00	80.00 $\pm$ 40.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Haircut Rate	Valid submission %	100	100	100	100
	Macro-F1 %	20.35 $\pm$ 0.00	11.01 $\pm$ 0.47	26.39 $\pm$ 0.00	13.97 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	80.00 $\pm$ 40.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Housekeeping	Valid submission %	40	100	0	40
	Macro-F1 %	62.14 $\pm$ 0.00	58.74 $\pm$ 0.00	–	65.05 $\pm$ 0.37
	Helpful selection %	100 $\pm$ 0.00	80.00 $\pm$ 40.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Machine Failure	Valid submission %	0	80	80	20
	Macro-F1 %	–	69.91 $\pm$ 0.01	80.32 $\pm$ 0.00	76.76 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	40.00 $\pm$ 54.77	100 $\pm$ 0.00	80.00 $\pm$ 40.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Song Popularity	Valid submission %	100	20	80	80
	Macro-F1 %	36.45 $\pm$ 0.16	30.46 $\pm$ 0.00	39.85 $\pm$ 15.60	40.23 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	80.00 $\pm$ 40.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Wine Quality	Valid submission %	100	100	100	100
	Macro-F1 %	25.43 $\pm$ 0.20	21.80 $\pm$ 0.11	27.33 $\pm$ 0.08	23.37 $\pm$ 0.25
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Game Revenue	Valid submission %	80	20	80	80
	RMSE	65.46 $\pm$ 0.67	64.89 $\pm$ 0.00	58.93 $\pm$ 0.78	58.72 $\pm$ 0.21
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Power Generation	Valid submission %	0	60	100	0
	RMSE	–	526.23 $\pm$ 0.00	519.64 $\pm$ 0.00	–
	Helpful selection %	100 $\pm$ 0.00	80.00 $\pm$ 40.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Real Estate	Valid submission %	0	100	0	100
	RMSE	–	58.25 $\pm$ 0.00	–	51.10 $\pm$ 0.51
	Helpful selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00
Second-hand Goods	Valid submission %	40	0	20	40
	RMSE	3.47 $\pm$ 0.03	–	3.36 $\pm$ 0.00	3.53 $\pm$ 0.00
	Helpful selection %	100 $\pm$ 0.00	80.00 $\pm$ 40.00	100 $\pm$ 0.00	100 $\pm$ 0.00
	Adversarial selection %	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00	100 $\pm$ 0.00

Table 17: Selection rates of adversarial notebooks by GPT-4o-mini across different datasets with varying helpful to adversarial notebook ratios. The table shows consistent preference for adversarial notebooks (nearly 100% selection rate) regardless of dataset or notebook ratio, with minimal selection of helpful notebooks.

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	100.0	80.0	80.0
bike-sharing-demand	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	6.7 $\pm$ 13.3
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	40.0	0.0	40.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	40.0	20.0	0.0
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	6.7 $\pm$ 13.3
	Adversarial selection%	100.0 $\pm$ 0.0	93.3 $\pm$ 13.3	100.0 $\pm$ 0.0
	Valid submission%	0.0	0.0	0.0
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	0.0	0.0	0.0

Table 18: Selection rates of adversarial notebooks by GPT-4o across different datasets with varying helpful to adversarial notebook ratios. The table shows consistent preference for adversarial notebooks (nearly 100% selection rate) regardless of dataset or notebook ratio, with minimal selection of helpful notebooks.

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	93.3 $\pm$ 13.3	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	80.0	100.0	80.0
bike-sharing-demand	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	13.3 $\pm$ 26.7
	Adversarial selection%	66.7 $\pm$ 21.1	73.3 $\pm$ 13.3	100.0 $\pm$ 0.0
	Valid submission%	40.0	60.0	20.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	20.0	20.0	20.0
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	0.0	0.0	0.0
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	0.0	0.0	0.0

Table 19: Selection rates of adversarial notebooks by GPT-o4-mini across different datasets with varying helpful to adversarial notebook ratios. The table shows consistent preference for adversarial notebooks (nearly 100% selection rate) regardless of dataset or notebook ratio, with minimal selection of helpful notebooks.

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	80.0	40.0	100.0
bike-sharing-demand	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	46.7 $\pm$ 26.7	73.3 $\pm$ 32.7	100.0 $\pm$ 0.0
	Valid submission%	60.0	80.0	40.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	60.0 $\pm$ 32.7	73.3 $\pm$ 32.7	100.0 $\pm$ 0.0
	Valid submission%	100.0	100.0	80.0
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	20.0	20.0	20.0
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Valid submission%	0.0	20.0	20.0

Table 20: Kaggle dataset results with temperature = 0.7, GPT-4o-mini and high_1, high_2, high_3 (Task 1).

Dataset	Metric	No. of Helpful Documents		
		1	2	3
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	86.7 $\pm$ 16.3
	Adversarial selection%			
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	73.3 $\pm$ 13.3
	Adversarial selection%			
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%			
otto-group-product-classification-challenge	Helpful selection%	60.0 $\pm$ 49.0	50.0 $\pm$ 0.0	66.7 $\pm$ 21.1
	Adversarial selection%			
rossmann-store-sales	Helpful selection%	20.0 $\pm$ 40.0	100.0 $\pm$ 0.0	93.3 $\pm$ 13.3
	Adversarial selection%			

Table 21: Kaggle dataset results with temperature = 0.7, GPT-4o and high_1, high_2, high_3 (Task 1).

Dataset	Metric	No. of Helpful Documents		
		1	2	3
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	60.0 $\pm$ 20.0	66.7 $\pm$ 21.1
	Adversarial selection%			
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	70.0 $\pm$ 24.5	33.3 $\pm$ 0.0
	Adversarial selection%			
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%			
otto-group-product-classification-challenge	Helpful selection%	100.0 $\pm$ 0.0	60.0 $\pm$ 20.0	66.7 $\pm$ 21.1
	Adversarial selection%			
rossmann-store-sales	Helpful selection%	100.0 $\pm$ 0.0	50.0 $\pm$ 0.0	66.7 $\pm$ 21.1
	Adversarial selection%			

Table 22: Kaggle dataset results with GPT-o4-mini and high_1, high_2, high_3 (Task 1).

Dataset	Metric	No. of Helpful Documents		
		1	2	3
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
	Adversarial selection%			
bike-sharing-demand	Helpful selection%	80.0 $\pm$ 40.0	50.0 $\pm$ 0.0	40.0 $\pm$ 13.3
	Adversarial selection%			
bnp-paribas-cardif-claims-management	Helpful selection%	20.0 $\pm$ 40.0	40.0 $\pm$ 20.0	66.7 $\pm$ 21.1
	Adversarial selection%			
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	20.0 $\pm$ 24.5	6.7 $\pm$ 13.3
	Adversarial selection%			
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	20.0 $\pm$ 16.3
	Adversarial selection%			

Table 23: Kaggle dataset results with temperature = 0.7, GPT-4o-mini and high_1_low_3, high_2_low_3, high_3_low_1 (Task 2).

Dataset	Metric	Helpful : Low		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	70.0 $\pm$ 24.5	80.0 $\pm$ 16.3
	Adversarial selection%			
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	66.7 $\pm$ 0.0
	Adversarial selection%			
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%			
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	50.0 $\pm$ 44.7	26.7 $\pm$ 32.7
	Adversarial selection%			
rossmann-store-sales	Helpful selection%	100.0 $\pm$ 0.0	60.0 $\pm$ 49.0	93.3 $\pm$ 13.3
	Adversarial selection%			

Table 24: Kaggle dataset results with temperature = 0.7, GPT-4o and high_1_low_3, high_2_low_3, high_3_low_1 (Task 2).

Dataset	Metric	Helpful : Low		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	80.0 $\pm$ 40.0	70.0 $\pm$ 24.5	66.7 $\pm$ 0.0
	Adversarial selection%			
bike-sharing-demand	Helpful selection%	20.0 $\pm$ 40.0	10.0 $\pm$ 20.0	60.0 $\pm$ 13.3
	Adversarial selection%			
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%			
otto-group-product-classification-challenge	Helpful selection%	20.0 $\pm$ 40.0	10.0 $\pm$ 20.0	53.3 $\pm$ 16.3
	Adversarial selection%			
rossmann-store-sales	Helpful selection%	20.0 $\pm$ 40.0	20.0 $\pm$ 24.5	46.7 $\pm$ 16.3
	Adversarial selection%			

Table 25: Kaggle dataset results with GPT-o4-mini and high_1_low_3, high_2_low_3, high_3_low_1 (Task 2).

Dataset	Metric	Helpful : Low		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
	Adversarial selection%			
bike-sharing-demand	Helpful selection%	80.0 $\pm$ 40.0	60.0 $\pm$ 20.0	33.3 $\pm$ 0.0
	Adversarial selection%			
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	50.0 $\pm$ 31.6	40.0 $\pm$ 13.3
	Adversarial selection%			
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	10.0 $\pm$ 20.0	0.0 $\pm$ 0.0
	Adversarial selection%			
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	10.0 $\pm$ 20.0	0.0 $\pm$ 0.0
	Adversarial selection%			

Table 26: Kaggle dataset results with temperature = 0.7, GPT-4o-mini and high_1_adv_3_neu_prompt, high_2_adv_3_neu_prompt, high_3_adv_1_neu_prompt (Task 3).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	50.0 $\pm$ 0.0	86.7 $\pm$ 16.3
	Adversarial selection%	80.0 $\pm$ 26.7	53.3 $\pm$ 16.3	80.0 $\pm$ 40.0
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	66.7 $\pm$ 0.0
	Adversarial selection%	60.0 $\pm$ 13.3	53.3 $\pm$ 26.7	100.0 $\pm$ 0.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	93.3 $\pm$ 13.3	86.7 $\pm$ 26.7	100.0 $\pm$ 0.0
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	40.0 $\pm$ 20.0	93.3 $\pm$ 13.3
	Adversarial selection%	93.3 $\pm$ 13.3	13.3 $\pm$ 16.3	60.0 $\pm$ 49.0
rossmann-store-sales	Helpful selection%	60.0 $\pm$ 49.0	0.0 $\pm$ 0.0	40.0 $\pm$ 24.9
	Adversarial selection%	73.3 $\pm$ 38.9	93.3 $\pm$ 13.3	80.0 $\pm$ 40.0

Table 27: Kaggle dataset results with temperature = 0.7, GPT-4o and high_1_adv_3_neu_prompt, high_2_adv_3_neu_prompt, high_3_adv_1_neu_prompt (Task 3).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	0.0 $\pm$ 0.0	20.0 $\pm$ 24.5	73.3 $\pm$ 24.9
	Adversarial selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	20.0 $\pm$ 40.0
bike-sharing-demand	Helpful selection%	80.0 $\pm$ 40.0	50.0 $\pm$ 31.6	60.0 $\pm$ 24.9
	Adversarial selection%	73.3 $\pm$ 32.7	60.0 $\pm$ 32.7	100.0 $\pm$ 0.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	93.3 $\pm$ 13.3	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	13.3 $\pm$ 16.3
	Adversarial selection%	93.3 $\pm$ 13.3	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	100.0 $\pm$ 0.0	73.3 $\pm$ 24.9	100.0 $\pm$ 0.0

Table 28: Kaggle dataset results with GPT-o4-mini and high_1_adv_3_neu_prompt, high_2_adv_3_neu_prompt, high_3_adv_1_neu_prompt (Task 3).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bike-sharing-demand	Helpful selection%	80.0 $\pm$ 40.0	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
	Adversarial selection%	6.7 $\pm$ 13.3	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	10.0 $\pm$ 20.0	20.0 $\pm$ 26.7
	Adversarial selection%	60.0 $\pm$ 38.9	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	10.0 $\pm$ 20.0	0.0 $\pm$ 0.0
	Adversarial selection%	20.0 $\pm$ 40.0	6.7 $\pm$ 13.3	80.0 $\pm$ 40.0
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	20.0 $\pm$ 24.5	6.7 $\pm$ 13.3
	Adversarial selection%	40.0 $\pm$ 49.0	40.0 $\pm$ 49.0	60.0 $\pm$ 49.0

Table 29: Kaggle dataset results with temperature = 0.7, GPT-4o-mini and high_1_adv_3_helpful_prompt, high_2_adv_3_helpful_prompt, high_3_adv_1_helpful_prompt (Task 4).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Adversarial selection%	0.0 $\pm$ 0.0	33.3 $\pm$ 0.0	0.0 $\pm$ 0.0
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	80.0 $\pm$ 16.3
	Adversarial selection%	13.3 $\pm$ 26.7	33.3 $\pm$ 0.0	40.0 $\pm$ 49.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	93.3 $\pm$ 13.3	86.7 $\pm$ 16.3	100.0 $\pm$ 0.0
otto-group-product-classification-challenge	Helpful selection%	0.0 $\pm$ 0.0	50.0 $\pm$ 31.6	66.7 $\pm$ 0.0
	Adversarial selection%	46.7 $\pm$ 45.2	26.7 $\pm$ 13.3	100.0 $\pm$ 0.0
rossmann-store-sales	Helpful selection%	100.0 $\pm$ 0.0	20.0 $\pm$ 40.0	60.0 $\pm$ 32.7
	Adversarial selection%	60.0 $\pm$ 32.7	80.0 $\pm$ 26.7	80.0 $\pm$ 40.0

Table 30: Kaggle dataset results with temperature = 0.7, GPT-4o and high_1_adv_3_helpful_prompt, high_2_adv_3_helpful_prompt, high_3_adv_1_helpful_prompt (Task 4).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	53.3 $\pm$ 16.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	86.7 $\pm$ 16.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	40.0 $\pm$ 49.0
otto-group-product-classification-challenge	Helpful selection%	100.0 $\pm$ 0.0	30.0 $\pm$ 24.5	73.3 $\pm$ 13.3
	Adversarial selection%	0.0 $\pm$ 0.0	33.3 $\pm$ 36.5	80.0 $\pm$ 40.0
rossmann-store-sales	Helpful selection%	20.0 $\pm$ 40.0	40.0 $\pm$ 37.4	26.7 $\pm$ 24.9
	Adversarial selection%	66.7 $\pm$ 42.2	40.0 $\pm$ 49.0	100.0 $\pm$ 0.0

Table 31: Kaggle dataset results with GPT-o4-mini and high_1_adv_3_helpful_prompt, high_2_adv_3_helpful_prompt, high_3_adv_1_helpful_prompt (Task 4).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	70.0 $\pm$ 24.5	40.0 $\pm$ 13.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	40.0 $\pm$ 13.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bnp-paribas-cardif-claims-management	Helpful selection%	60.0 $\pm$ 49.0	90.0 $\pm$ 20.0	33.3 $\pm$ 29.8
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	60.0 $\pm$ 49.0
otto-group-product-classification-challenge	Helpful selection%	40.0 $\pm$ 49.0	40.0 $\pm$ 37.4	26.7 $\pm$ 24.9
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	20.0 $\pm$ 40.0
rossmann-store-sales	Helpful selection%	0.0 $\pm$ 0.0	40.0 $\pm$ 37.4	40.0 $\pm$ 24.9
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0

Table 32: Kaggle dataset results with temperature = 0.7, GPT-4o-mini and high_1_adv_3_specific_helpful_prompt, high_2_adv_3_specific_helpful_prompt, high_3_adv_1_specific_helpful_prompt (Task 5).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	93.3 $\pm$ 13.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Adversarial selection%	0.0 $\pm$ 0.0	33.3 $\pm$ 21.1	40.0 $\pm$ 49.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	20.0 $\pm$ 40.0	20.0 $\pm$ 26.7	0.0 $\pm$ 0.0
otto-group-product-classification-challenge	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	40.0 $\pm$ 49.0
rossmann-store-sales	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	Adversarial selection%	13.3 $\pm$ 26.7	46.7 $\pm$ 26.7	20.0 $\pm$ 40.0

Table 33: Kaggle dataset results with temperature = 0.7, GPT-4o and high_1_adv_3_specific_helpful_prompt, high_2_adv_3_specific_helpful_prompt, high_3_adv_1_specific_helpful_prompt (Task 5).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	80.0 $\pm$ 26.7
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	93.3 $\pm$ 13.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	20.0 $\pm$ 40.0
bnp-paribas-cardif-claims-management	Helpful selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
otto-group-product-classification-challenge	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	60.0 $\pm$ 24.9
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
rossmann-store-sales	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	66.7 $\pm$ 21.1
	Adversarial selection%	20.0 $\pm$ 40.0	0.0 $\pm$ 0.0	60.0 $\pm$ 49.0

Table 34: Kaggle dataset results with GPT-o4-mini and high_1_adv_3_specific_helpful_prompt, high_2_adv_3_specific_helpful_prompt, high_3_adv_1_specific_helpful_prompt (Task 5).

Dataset	Metric	Helpful : Adversarial		
		1:3	2:3	3:1
allstate-claims-severity	Helpful selection%	100.0 $\pm$ 0.0	80.0 $\pm$ 24.5	46.7 $\pm$ 16.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bike-sharing-demand	Helpful selection%	100.0 $\pm$ 0.0	50.0 $\pm$ 0.0	53.3 $\pm$ 26.7
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
bnp-paribas-cardif-claims-management	Helpful selection%	80.0 $\pm$ 40.0	90.0 $\pm$ 20.0	66.7 $\pm$ 29.8
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
otto-group-product-classification-challenge	Helpful selection%	100.0 $\pm$ 0.0	60.0 $\pm$ 37.4	26.7 $\pm$ 13.3
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
rossmann-store-sales	Helpful selection%	80.0 $\pm$ 40.0	50.0 $\pm$ 31.6	33.3 $\pm$ 21.1
	Adversarial selection%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0